

CS201 Midterm 3

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Abstract—Write only ONE function per question in C++. Do not check the input, assume that the input is correct. Unless noted, (i) the input data structure is not empty, (ii) time complexity of the algorithm does not matter, (iii) you can not use extra data structures, (iv) you can use any function given in the data structure.

I. QUESTION (HEAP), 20 POINTS

Write the method in **MinHeap** class

```
bool isLargestLeftMost()
```

which returns true if the largest value appears on the leftmost node. You may not use any additional data structures.

II. QUESTION (HEAP), 15 POINTS

Write the method in **MaxDHeap** class

```
int howManyPairCanBeSwapped()
```

that returns the number of node pairs in the heap which can be swapped without hurting the heap property. Your method should run in $O(N^2)$ time. You may not use any additional data structures.

III. QUESTION (DISJOINT SET), 15 POINTS

Write the method in **DisjointSet** class

```
int* grandchildren(int index)
```

where given the index of a set, it returns the indexes of its grandchildren as an array. Do not use any class or external methods.

IV. QUESTION (DISJOINT SET), 15 POINTS

Write the method in **DisjointSet** class

```
int numberOfTriplets()
```

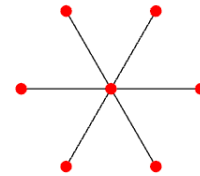
that calculates the number of triplet disjoint sets in a disjoint set structure. A disjoint set is a triplet, if the number of sets in that disjoint set is 3. Do not use any class or external methods.

V. QUESTION (GRAPH), 20 POINTS

Write the method in array (adjacency matrix) implementation

```
bool isStarGraph()
```

which returns whether graph is a star graph or not. Star graph is obtained by connecting a node to all the remaining nodes. If a graph has n nodes, there are $n-1$ edges as shown in example star graph below. You are not allowed to use depth first search or breadth first search.



VI. QUESTION (GRAPH), 15 POINTS

Write the method in linked list implementation

```
Graph* inverseGraph()
```

which constructs an inverse graph of a given graph in linked list implementation. In inverse graph, two distinct vertices are adjacent if and only if they are not adjacent in the original graph. You are not allowed to use extra data structures apart from the constructed graph.